Damian L. Gallegos Espinoza

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Research Summary

Ph.D. student working on **trustworthy machine learning**: quantifying uncertainty and estimating error in neural networks, with a focus on **Kolmogorov–Arnold Networks (KANs)**.

Education

Ph.D. in Computer Science, The University of Texas at El Paso (GPA 4.00) Jan. 2025 – Present

M.S. in Data and Information Science, The University of Texas at El Paso (GPA 3.72) Aug. 2022 – Dec. 2024

Thesis: Machine learning for water-use prediction.

B.S. in Materials Engineering, Autonomous University of Ciudad Juárez (GPA 3.88) Aug. 2017 – May 2021

Research Experience

Doctoral Researcher | *The University of Texas at El Paso* Jan. 2025 – Present

- Proposed adaptive optimizer sequencing for KANs (L-BFGS first), validated by ablations vs. baselines [1].
- Released open-source code with a reproducible experiment-tracking workflow.
- Turning trained KANs into readable mathematical formulas (Julia) to make models explainable.
- Developing methods to estimate model confidence and error, including Hessian-based analysis.

Graduate Research Assistant | *The University of Texas at El Paso* Jan. 2024 – Aug. 2024

- Built deep learning models to predict urban water use; led to two MICAI publications and my M.S. thesis.

Publications

[1] **D. L. Gallegos Espinoza**, Y. I. Guel, M. N. Sojib, M. Ceberio. **Adaptive Optimizer Sequencing for Kolmogorov–Arnold Networks.** *Advances in Soft Computing (MICAI 2026)*, LNAI vol. 17149, Springer, 2027. [DOI] [Code]

[2] M. N. Sojib, **D. L. Gallegos Espinoza**, M. Ceberio. **FedAWL: Client Selection via Age-Weighted Loss for Non-IID Federated Learning.** *Advances in Soft Computing (MICAI 2026)*, LNAI vol. 17149, Springer, 2027. [DOI]

[3] **D. L. Gallegos Espinoza**, O. Kosheleva, V. Kreinovich. **Why Skew-Normal Distributions and How They Are Related to ReLU Activation Function in Deep Learning.** *Data Science in Finance and Accounting*, Studies in Big Data vol. 181, Springer, 2026. [DOI]

Industry Experience

Data Science Intern | *North American Development Bank* May 2024 – Aug. 2024

- Built a Python system fusing climate, financial, and geospatial data to flag project risk early.
- Automated managers' manual project-status reporting (80% less time) with Power BI dashboards.

Test Engineer | *Foxconn* Dec. 2021 – Aug. 2022

- Automated server testing and failure analysis with Python and Bash tools.

Teaching Experience

Graduate Teaching Assistant | *The University of Texas at El Paso* 2023 – Present

CS 4381/5381	Cross-Platform App Development *(Dart, Flutter)*	Summer, Fall 2026
CS 4342/5342	Database Management *(SQL, relational design)*	Fall 2024 – Spring 2026
CS 2401	Data Structures and Algorithms *(Java)*	Spring, Fall 2023

Skills

ML & Statistics	Uncertainty quantification, probabilistic modeling, interpretable/symbolic ML, federated learning
Optimization	Second-order methods (L-BFGS), Hessian analysis, optimizer scheduling
Experimentation	Ablations, baselines and metrics, reproducible pipelines, open-source research code
Sci. Computing	Python (PyTorch, scikit-learn, NumPy), Julia
Software	Java, SQL, Dart/Flutter, Git, Linux, LaTeX